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Light curves sequential comparison strategy for improved understanding of LEO uncontrolled objects

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ABSTRACT

In the framework of Space Situational Awareness activities, space debris light curve analyses have been used as a tool to study the tumbling motion of uncontrolled orbiting objects using only optical data and often without any assumptions about their geometry. Through this strategy, frequency analysis can be performed to extract the objects' rotation periods, which are necessary to predict variations in attitude for both monitoring and management purposes. In the context of international tasks and activities, the Sapienza Space System and Space Surveillance Laboratory research group at Sapienza University of Rome, on behalf of the Italian Space Agency delegation to the Inter-Agency Space Debris Coordination Committee, has created a historical record of light curve data by observing seven rocket bodies in low Earth orbit. This dataset, supplemented with further investigation over the years, has revealed the long-term evolution of the motion states, with particular attention to the recorded tumbling rates, which would be unobservable with short-term campaigns. These data allow for long-term light curve inversion and historical analyses of rotational frequency and photometric response to optical observations, which are of interest to Space Surveillance and Tracking and Space Situational Awareness services for database updates, monitoring of uncontrolled resident space objects, and future re-entry observations. In this paper, the sequential comparison strategy used to investigate the long-term evolution of space debris rotation rates is presented, with a specific focus on the SL-14 R/B (NORAD ID 18340), which has exhibited an anomalous increase in its rotation period over the years. To investigate the underlying causes of this behaviour, a dedicated attitude evolution analysis was conducted, employing various methodologies. First, attitude simulation techniques were used to explore potential root causes of this sudden acceleration, including the hypothesis of a failure in the propulsion system leading to pressurized gas leakage. Different leakage models were analysed to assess the plausibility of this scenario. Additionally, a digital twin simulation was developed to perform light curve inversion, enabling the reconstruction and validation of the object's attitude motion. The results of these analyses provide deeper insight into the dynamics of non-cooperative space objects, fostering the development of improved tracking, characterization, and predictive modelling strategies in support of Space Surveillance and Tracking activities.

1. Introduction

The exponential increase in space debris in recent decades has led to a growing concern within the international space community. Space agencies and private companies operating in the space sector have recognized the need for improved Space Surveillance and Tracking (SST) capabilities to mitigate collision risks and ensure the sustainability of space activities. According to the European Space Agency (ESA), approximately 40,000 space debris larger than 10 cm are currently in Earth's orbit, along with millions of smaller objects [1]. These uncontrolled objects, resulting from satellite breakups, defunct spacecraft, and spent rocket stages, pose a significant threat to operational satellites and human missions. In particular, collisions with space debris can have

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Acronyms/abbreviations

AIUB – Astronomical Institute of the University of Bern
ASI – Italian Space Agency
CAD – Computer Aided Design
Dec – Declination
ESA – European Space Agency
GG – Gravity Gradient
IADC – Inter-Agency Space Debris Coordination Committee
JAXA – Japan Aerospace Exploration Agency
LC – Light Curve
LEO – Low Earth Orbit
MOGA – Multi- Objective Genetic Algorithm
NASA – National Aeronautics and Space Administration
NORAD – North American Aerospace Defence Command
NTO – Nitrogen tetroxide
R/B – Rocket Body
RA – Right Ascension
RSO – Resident Space Object
S5Lab – Sapienza Space System and Space Surveillance Laboratory
SCUDO – Sapienza Coupled University Debris Observatory
SRP – Solar Radiation Pressure
SSA – Space Situational Awareness
SSAU – State Space Agency of Ukraine
SST – Space Surveillance and Tracking
TDM – Tracking Data Message
TLE – Two Line Elements
UDMH – Unsymmetrical dimethylhydrazine
VISONE – VIrtual Space ObservatioN Engine

catastrophic consequences, potentially triggering cascading fragmentation events, commonly referred to as the Kessler Syndrome [2]. To address these challenges, international efforts have been directed toward developing advanced techniques for tracking, monitoring, and characterizing space debris. A significant aspect of space debris monitoring involves understanding its rotational behavior, which can provide useful improvements in the general characterization of a Resident Space Object's (RSO) dynamical state [3]. In fact, the attitude state of a RSO, defined by its rotational motion, is strongly coupled with its orbital dynamics. Therefore, precise knowledge of a RSO's attitude is essential for improving orbit determination and further analysis depending on orbit propagations [4]. One of the most employed techniques used for studying the rotational behavior of space debris is light curve analysis [5]. A light curve (LC) represents the variation in a RSO's brightness over time, typically measured in magnitudes. The periodicity of brightness behavior can be correlated with specific rotational or tumbling motions, providing an effective method for estimating a RSO's spin rate, stability, and potential perturbations affecting its motion. Light curves can be obtained from optical observations. Previous studies have demonstrated that RSOs experiencing uncontrolled tumbling motion tend to exhibit periodic variations in their light curves [6]. By applying frequency-domain analysis techniques, such as Fast Fourier Transform (FFT), Lomb Scargle periodogram and Phase Dispersion Minimization (PDM), one can extract the dominant periodicities from the light curve data, enabling an estimate of a RSO's attitude evolution over time [7]. This study focuses on the SL-14 R/B (NORAD ID 18340), a spent rocket body that has been the subject of a long-term observational campaign conducted under the Inter-Agency Space Debris Coordination Committee (IADC) framework. The IADC is an international committee that promotes coordinated research efforts aimed at characterizing and mitigating the space debris problem. Observations of SL-14 R/B have been carried out by multiple space agencies, including ASI (Italian Space Agency), through the Sapienza Space Systems and Space Surveillance Laboratory (S5Lab) at Sapienza University of Rome. Optical data acquired from 2020 to the present have provided a unique opportunity to analyze the RSO's rotational behavior over an extended period. Analysis of the light curve data for SL-14 R/B has revealed an unexpected acceleration in its rotation period. Typically, non-cooperative RSOs such as space debris tend to stabilize their tumbling motion over time due to energy dissipation effects, including eddy current damping and internal energy dissipation. However, in this case, a sudden and significant increase in rotational rate has been detected. This anomaly has raised considerable interest, and consequently, a study of the causes of this sudden acceleration has been conducted, developing an attitude propagation model that accounts for the main perturbations acting on SL-14 R/B's orbit, a LEO orbit with an altitude ranging from 1417 km to 1479 km and an inclination of 82°. The study hypothesizes that the acceleration is caused by a thrust resulting from the release of pressurized gas still present in the rocket body's tanks. Different leakage assumptions were developed, to fit analytical models with observed data and studying the results compatibility with the rocket body's engine features. Finally, computer simulations aimed at virtually replicating the real optical observations were performed using SL-14 R/B's digital twin.

This paper is structured as follows. The first section presents the results of the observation campaign of Low Earth Orbit (LEO) R/Bs, focusing on LC analysis and sequential comparison methodologies used to track the long-term evolution of their motion states. Particular attention is given to the SL-14 R/B (NORAD ID 18340), which has exhibited anomalous variations in its tumbling rate over time. The second section is dedicated to attitude simulations aimed at investigating the possible causes of this anomalous behaviour, exploring different hypotheses such as external perturbations and internal failures, including potential pressurized gas leakage. Finally, the third section introduces the use of a digital twin approach for light curve inversion, enabling the reconstruction of the RSO's attitude evolution during a pass over the observatory. This methodology enhances the understanding of the SL-14 R/B's rotational dynamics and provides a more comprehensive assessment of its motion state.

2. Light curve sequential comparison strategy

From September 2020 to January 2022, the S5Lab research team participated in the IADC WG1 AI-38.2 observation campaign titled "*Attitude Motion Characterization of LEO Upper Stages using Different Observation Techniques*" [8]. This campaign focused on monitoring seven LEO rocket bodies, including the Russian SL-8, SL-14, and SL-16 R/B, the Chinese CZ-4C and CZ-11 R/B, and the American Pegasus R/B. It resulted in a comprehensive dataset of LC data, organised into standardised data-sharing files known as Tracking Data Message (TDM). These files allowed the comparison with other IADC member space agencies. Three types of TDM files were generated: one containing Right Ascension (RA) and Declination (Dec), one with elevation and range, and another detailing magnitude and magnitude error. The results of the observation campaign are summarised in Table 1.

The collected data were analysed to detect the presence of tumbling RSOs by studying the periodic variations in brightness. These variations correspond directly to the satellite's rotation rate or precession period. A flat spinning tumbling motion was detected for the SL-14 R/B. As shown in Fig. 1, its LC exhibits a periodic pattern with fluctuations ranging from magnitudes 7 to 10 and a period of 88.5 s. From the figure are evident different local minima and maxima values indicating that the object is rotating along different axes, however the periodicity in the light curve pattern indicates a stable rotation.

Once the tumbling motion was detected and confirmed for SL-14 R/

Table 1
Dataset acquired during the WG1 AI 38.2 observation campaign (September 2020–January 2022).

NORAD ID	Observations	Light Curve	TDM
11804	10	10	30
18340	20	28	84
23088	2	4	12
23405	10	12	26
39198	9	9	27
40879	7	7	21
41847	21	23	69
TOTAL	**79**	**93**	**279**

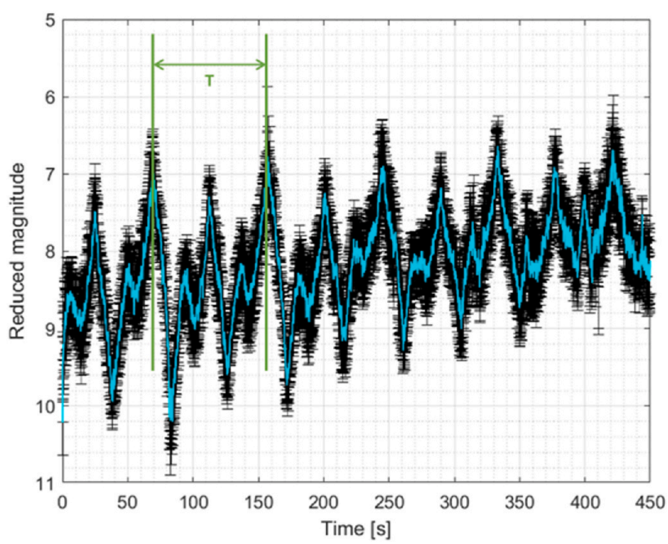

Fig. 1. Observed normalised LC of SL-14 R/B (NORAD ID 18340) acquired from SCUDO on July 29, 2021.

B, follow-up observation activities were planned to track its evolution. To fully understand a SL-14 R/B's attitude and its change over the years, the so-called "Light Curve Sequential Comparison Strategy" was applied. This strategy consists of periodically observing the same targets during consecutive observation nights, to create a sequence of LC data points spaced in time. The data can then be compared with the attitude evolution predicted by mathematical models, specifically the rigid-body dynamics model incorporating gravity gradient and eddy current torques. These perturbation effects were used to simulate the natural evolution of the tumbling motion, providing a reference against which the observed light curve data could be analysed.

. This approach allows rapid detection of any changes in the attitude motion and the evolution of the tumbling states, as it captures the RSO's behaviour over an extended period. By applying this strategy to SL-14 R/B, an anomalous abrupt acceleration was discovered starting in July 2022. This acceleration was confirmed by the members of the IADC WG1, further validating the findings. By combining the acquired data from all the participants, it was possible to produce the graph shown in Fig. 2, where blue markers correspond to ESA data, cyan to Japan Aerospace Exploration Agency (JAXA), yellow to the State Space Agency of Ukraine (SSAU), red to the Astronomical Institute of the University of Bern (AIUB) and bordeaux to S5Lab [9].

From the analysis of the data shown in Fig. 2, three distinct phases in the evolution of the tumbling motion of SL-14 R/B can be identified.

1) **Phase 1 (September 2020 – July 2022):** The SL-14 R/B exhibits nominal tumbling motion typical of an uncontrolled rocket body. During this period, a gradual increase in the apparent rotation period is observed, consistent with the dissipation of kinetic energy due to internal and external torques.
2) **Phase 2 (July 2022 – February 2024)**: A sudden acceleration in the SL-14 R/B's rotation is detected, marking a deviation from the expected motion. The end of this acceleration phase is estimated to have occurred around February 2024, based on the currently available data. However, if additional observational data from other agencies were to become available, a more precise determination of the transition point could be achieved.
3) **Phase 3 (February 2024 – January 2025)**: Following the acceleration, the SL-14 R/B enters a deceleration phase, characterised by an increase in the apparent rotation period, suggesting a loss of angular momentum. This phase marks a return to a more dissipative state, similar to the pre-acceleration period.

3. Attitude simulations of SL-14 R/B (NORAD ID 18340)

To expand this research beyond simple tumbling detection, further investigations were carried out to identify the underlying causes of the changes in SL-14 R/B's behaviour, with a focus on its shape, orbit, and

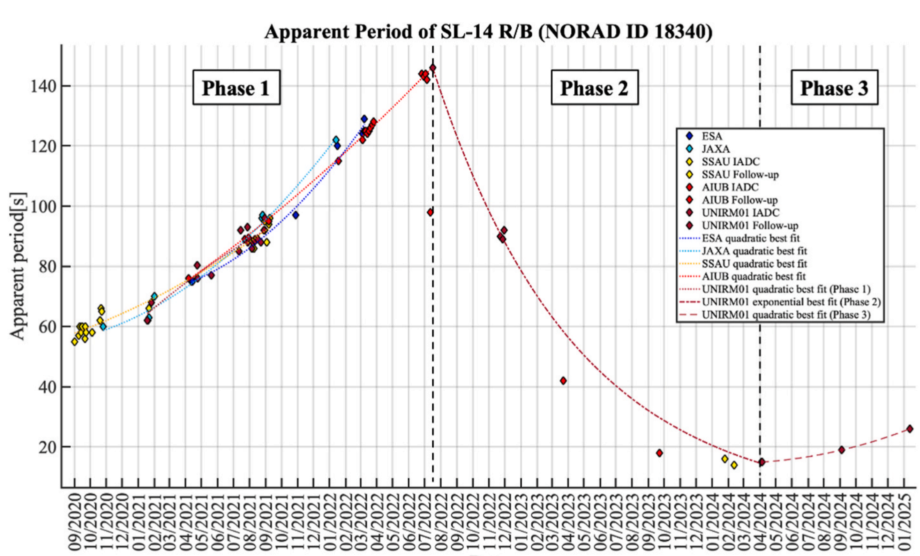

Fig. 2. Apparent period evolution of SL-14 R/B (NORAD ID 18340) from September 2020 to January 2025. The data presented in Phase 1 have been extracted from the IADC WG1 AI38.2 graphs in Ref. [9].

attitude. Several hypotheses were considered to explain the sudden acceleration of the SL-14 R/B, including external impacts and internal mechanical failures. However, external impacts such as collisions with space debris or meteorites were ruled out, as they would have caused an immediate change in apparent rotation period followed by an immediate energy dissipation. After an extensive analysis of the observed data, the leading hypothesis was identified as a mechanical failure in the SL-14 R/B attitude control thrusters, specifically a depressurization event in the propulsion system (e.g., a failure in a valve), resulting in a pressurized gas leakage. This leakage could explain the abrupt increase in the SL-14 R/B tumbling motion, leading to the observed sudden decrease in its apparent period.

To validate this hypothesis, multiple attitude simulations were performed to correlate the SL-14 R/B's simulated attitude dynamics with the observed optical data. The first simulation, covering the period from September 2020 to July 2022, modelled the SL-14 R/B's attitude motion considering only natural environmental perturbations, specifically the gravity gradient and eddy current effects. Solar radiation pressure (SRP) perturbations were excluded, as the SRP torque is negligible compared to the effects of gravity gradient and eddy currents and does not significantly influence long-period attitude behaviour [10]. A second simulation, spanning from July 2022 to February 2024, was conducted to investigate the potential gas leakage from the attitude control thrusters. In this simulation, two different thrust models were implemented, and their validity was assessed by comparing the simulated apparent periods with the observed data. This analysis aimed to determine whether the pressurized gas leakage hypothesis could plausibly explain the observed tumbling behaviour. Finally, a comprehensive simulation was carried out, covering the period from September 2020 to January 2025. In this final simulation, the pressurized gas leakage disturbance was applied during the timespan from July 2022 to February 2024 to assess its long-term impact on the SL-14 R/B's tumbling motion.

3.1. Kinematics and dynamics models

The attitude representation chosen for the simulation is the quaternion, in the form $\boldsymbol{q} = [q_0, q_1, q_2, q_3]$ with q_0 scalar part. This representation avoids singularities in attitude propagation [11].

The kinematic is governed by Eq. (1):

$$\dot{\boldsymbol{q}}(t) = \frac{1}{2}\Omega(\boldsymbol{\omega}(t))\boldsymbol{q}(t) \tag{1}$$

In this formulation, $\boldsymbol{\omega}$ represents the SL-14 R/B's angular velocity in the body frame and the matrix $\Omega(\boldsymbol{\omega})$ is defined as shown in Eq. (2):

$$\Omega(\boldsymbol{\omega}) = \begin{bmatrix} -\boldsymbol{\omega}^T & 0 \\ -S(\boldsymbol{\omega}) & \boldsymbol{\omega} \end{bmatrix} \tag{2}$$

The dynamics of the debris is described using a rigid-body model and the state is propagated using Eq. (3):

$$\dot{\boldsymbol{\omega}}(t) = J^{-1}[\boldsymbol{T} - \boldsymbol{\omega}(t) \times J\boldsymbol{\omega}(t)] \tag{3}$$

In the previous equation, J is the inertia matrix in the body frame, while $\boldsymbol{T}$ comprises all the external torques. The inertia matrix value used in the attitude simulation is reported in Table 2. It was computed from the 3D Computer Aided Design (CAD) model of SL-14 R/B, which is shown in Fig. 3. The CAD model aims to reproduce its primary shapes, accounting for the original materials and dimensions, thus allowing for an accurate estimation of its inertia.

Table 2
Inertia matrix of SL-14 R/B (NORAD ID 18340) expressed in the body frame.

Inertia Matrix Component	Value $[\boldsymbol{kg{\cdot}m^2}]$
J_{xx}	$3.939{\cdot}10^3$
J_{yy}	$3.938{\cdot}10^3$
J_{zz}	$1.106{\cdot}10^3$
J_{xy}	$2.752{\cdot}10^{-2}$
J_{xz}	$-1.317{\cdot}10^{-1}$
J_{yz}	$3.879{\cdot}10^{-1}$

3.2. Perturbations

The behaviour of flat spinning debris in LEO is influenced by the gradual dissipation of rotational kinetic energy over time [12]. This dissipation results from various damping mechanisms that act to reduce the angular velocity of RSOs. The primary perturbations considered in attitude dynamics simulations are the gravity gradient and eddy currents. The gravity gradient exerts a torque due to the differential gravitational pull experienced by different parts of the RSO, while eddy currents are induced by the interaction of the debris with Earth's magnetic field, generating opposing magnetic forces that slow its rotation. These perturbative effects are incorporated into the simulations to accurately replicate the attitude evolution of the debris prior to any hypothetical failure of the thrust vector control system.

3.2.1. Gravity gradient

In LEO, a Gravitational Gradient (GG) is present due to the variation in gravitational pull across the length of the debris. This gradient exerts a stabilising torque on the RSO, aligning its minimum axis of inertia with the local vertical direction over time. This alignment reduces the tumbling motion and stabilizes the debris. The gravity gradient torque is computed according to the following Eq. (4) [13]:

$$T_{GG} = \frac{3\mu}{r^3}\boldsymbol{n} \times J^c\boldsymbol{n} \tag{4}$$

In the previous equation J^c is the inertia matrix of the RSO with respect to the centre of mass, $\boldsymbol{r}$ is the distance of the RSO from the centre of the Earth, $\boldsymbol{n}$ is the nadir unit vector expressed in the body frame, and μ is the Earth's gravitational parameter.

3.2.2. Eddy currents

The eddy currents have a significant effect on the attitude of tumbling RSOs [14]. These currents are induced by the Earth's magnetic field which interacts with conductive materials generating a torque. The eddy currents torque depends on the local magnetic field, on the physical characteristics of the material (mainly the electrical conductivity), and on the RSO's angular velocity. This perturbation is modelled according to the magnetic tensor formulation, detailed in Ref. [15]. The eddy currents perturbation for the rocket body is modelled using the

Fig. 3. 3D CAD model of SL-14 R/B (NORAD ID 18340).

magnetic tensor for a cylindrical RSO, resulting in a magnetic tensor as reported in Eq. (5):

$$M = \pi\sigma R^3 eL \begin{bmatrix} \gamma & 0 & 0 \\ 0 & \gamma & 0 \\ 0 & 0 & 0.5 \end{bmatrix} \tag{5}$$

where σ is the electrical conductivity of the material, R is the radius of the cylinder, L is the length of the cylinder, e is the thickness of the cylindrical shell, and $\gamma = 1 - \left(\frac{2R}{L}\right) tanh\left(\frac{L}{2R}\right)$. For the SL-14 R/B, the parameters selected are reported in Table 3. These values were chosen by considering the dimensions and materials of the rocket body. The value of the electrical conductivity has been selected considering that no official data on the Tsyklon-3 third stage structural material is available; therefore, it was assumed to be aluminium 6061-T4, a common aerospace alloy with a typical electrical conductivity of approximately 22×10^6 S/m [16].

The eddy currents torque is computed using Eq. (6) as follows:

$$T_{EC} = M(\boldsymbol{\omega} \times \boldsymbol{B}) \times \boldsymbol{B} \tag{6}$$

where $\boldsymbol{\omega}$ is the angular velocity of the engine, and $\boldsymbol{B}$ is the Earth's magnetic field vector, both in the body frame.

3.2.3. *Simulation results prior to hypothetical leakage*

The perturbation model was validated by simulating the attitude from September 2020 to July 2022, and by comparing the simulated apparent rotation periods with those obtained through observations. In this simulation, the only perturbations taken into account are the gravity gradient torque and eddy currents torque. The initial conditions for the angular velocity were derived from the observed apparent rotation period, $T_{app} = 55\,s$, as recorded during the September 2020 observations. Based on this period, the magnitude of the initial angular velocity was calculated using Eq. (7).

$$\omega = \frac{2\pi}{T_{app}} \tag{7}$$

The result of the simulation is presented in Fig. 4, which shows that the simulated apparent rotation period (curve) is compatible with the observed data (diamond markers), thus validating the perturbation model. The simulated periods are obtained directly from the norm of the angular velocity by inverting Eq. (7). It is important to note that this approach for computing the apparent period is not fully accurate, since the apparent period provided by optical observation depends also on the position of the observatory. Nevertheless, this procedure allows an overall assessment of the SL-14 R/B's tumbling behaviour.

3.3. *Leakage model*

Further research on SL-14 R/B has led to the identification of the RSO with the third stage of the Tsyklon-3 launcher [17] which featured the RD-861 engine as the propulsion system of the rocket's third stage. The RD-861, illustrated in Fig. 5, is a Soviet-era liquid-propellant rocket engine that burns unsymmetrical dimethylhydrazine (UDMH) and nitrogen tetroxide (NTO) in a gas-generator combustion cycle. It features a primary combustion chamber along with four nozzles for thrust vector control, powered by the gas generator output.

Table 3
Magnetic tensor coefficients of SL-14 R/B (NORAD ID 18340).

Property	Value
L	$3.15\,m$
σ	$22{\cdot}10^6\,S/m$
e	$0.001\,m$
R	$1.125\,m$

As previously mentioned, the leading hypothesis for explaining the SL-14 R/B sudden acceleration is a mechanical failure in one of the containment valves of the attitude thrusters, resulting in a leakage of the pressurant left in one of the propellant tanks. The presence of this disturbance was modelled in the simulations, starting with a trigonometric analysis of the CAD model, which, to a good approximation, provided the distances used to calculate the angular momentum generated by the thrust force and its intensity.

The thrust force is defined with magnitude m and direction indicated by unit vector $\hat{\boldsymbol{j}}$ in the body reference frame. The thrust direction $\hat{\boldsymbol{j}}$, is obtained from the unit vector $(-1, 0, 0)$ rotated in the XZ-plane – around the Y-axis – by $\pi/4$, as shown in Fig. 6.

The force equation is presented in Eq. (8), where the unit vector $\hat{\boldsymbol{j}}$ is multiplied by the leakage force magnitude:

$$\boldsymbol{J}(t) = m(t)\,\hat{\boldsymbol{j}} \tag{8}$$

The jet torque is given by Eq. (8):

$$\boldsymbol{T_J}(t) = \boldsymbol{b} \times \boldsymbol{J}(t) \tag{9}$$

where $\boldsymbol{b}$ is the vector from the centre of mass to the point where the jet force is applied.

In simulating the jet perturbation, it is assumed that only one nozzle is leaking. Two different thrust models were considered: an exponential thrust model, which represents the expected exhaust behaviour of a pressurized tank, and a continuous thrust model, which accounts for the potential failure of a mechanical valve within the system. To reproduce the sudden decrease of the apparent period, corresponding to an increase in the SL-14 R/B's angular velocity, the following environmental perturbations are combined with the jet leakage models.

3.3.1. *Exponential leakage*

The exponential model assumes a thrust intensity that decreases over time, that is a realistic representation of the exhaust behaviour in a pressurized tank. The adopted exponential decay model for the force is given by:

$$m(t) = \alpha{\cdot}\exp(-\beta \cdot t) \tag{10}$$

where α represents the initial thrust intensity, and β is the decay constant. As previously discussed, the thrust leakage is considered over the interval from July 6, 2022, to February 20, 2024, corresponding to a total thrust application duration of $51.408{\cdot}10^6$ seconds.

The first exponential model was designed to satisfy the condition that at the final epoch, the thrust intensity is:

$$m(t_{end}) = \alpha \cdot \exp(-7) \approx \alpha \cdot 9.1{\cdot}10^{-4} \tag{11}$$

which leads to a decay constant of $\beta = 1.36{\cdot}10^{-7}$. The simulated apparent period for $\beta = 1.36{\cdot}10^{-7}$ and two different values of α ($\alpha = 5.5{\cdot}10^{-4}\,N$ and $\alpha = 7.51{\cdot}10^{-4}\,N$) is shown in Fig. 7, and the corresponding thrust profiles are shown in Fig. 8. It is evident that this thrust model does not match the observed data: after an initial acceleration of the SL-14 R/B, the apparent period begins to increase again. This behaviour results from the rapid exponential decay of thrust intensity, combined with the eddy currents effect, governed by Eq. (6), becoming predominant as the angular velocity rises, thus causing an increase in the apparent period. Hence, to obtain a simulated apparent period compatible with the observations, a slower decay must be assumed for the trust.

A second exponential model was implemented with $\beta = 1.36{\cdot}10^{-9}$, resulting in a slower decay. In this case, the thrust intensity at the end of the simulation is 93 % of its initial value. As shown in. Fig. 9, the simulated apparent period obtained using this thrust model aligns well with the observational data, as expected. It is important to note that,

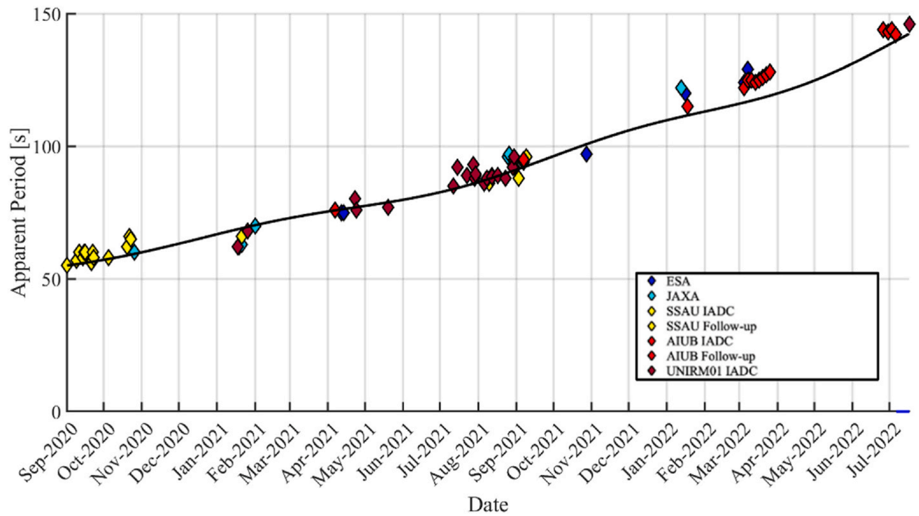

Fig. 4. SL-14 R/B (NORAD ID 18340) attitude simulation from September 2020 to July 2022 with gravity gradient and eddy currents perturbations.

Fig. 5. Tsyklon-3 third stage [18] and RD-861 engine [19].

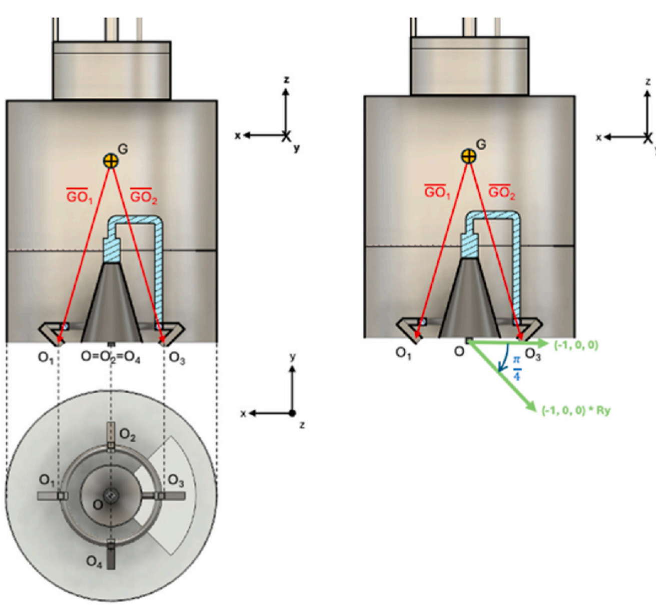

Fig. 6. Leakage jet vector calculation.

with the selected value of β, the thrust intensity does not follow a strictly exponential decay during the considered timespan. Instead, it behaves more like a linear decrease, as illustrated in Fig. 10, with a total variation of only 7 % between the initial and final values. These results highlight that achieving a simulated apparent period consistent with the observational data requires a higher total thrust impulse. Additionally, the nearly linear thrust profile observed in Fig. 10 supports the use of a continuous thrust model, as described in the next section.

3.3.2. *Continuous leakage*

The continuous model assumes a thrust that is constant over time, and different thrust values ($\alpha = 3.5{\cdot}10^{-4}\ N$, $\alpha = 5.5{\cdot}10^{-4}\ N$, and $\alpha = 7.5{\cdot}10^{-4}\ N$) were simulated as shown in Fig. 11. As expected, a higher thrust intensity leads to a faster acceleration of the SL-14 R/B than the exponential model. The continuous thrust model is plausible when considering a valve failure in one of the four engine nozzles. Since the engine is pressure-fed, once the liquid fuel is depleted, the gas will fill the tanks at a pressure equal to that required to pressurize the liquids. In the event of a mechanical failure in one of the vernier nozzle valves, a leakage of pressurized gas is expected. As the pressure in the system decreases, the valve would open wider, producing a nearly constant thrust and behaving like a pressure regulator. This behaviour would support the assumption of constant thrust.

Considering the RD-861 engine data [19], the thrust value was also determined. It was estimated that the total propellant mass is 3416 kg, with 1020 kg of UDMH and 1240 kg of NTO, corresponding to 1290 L of

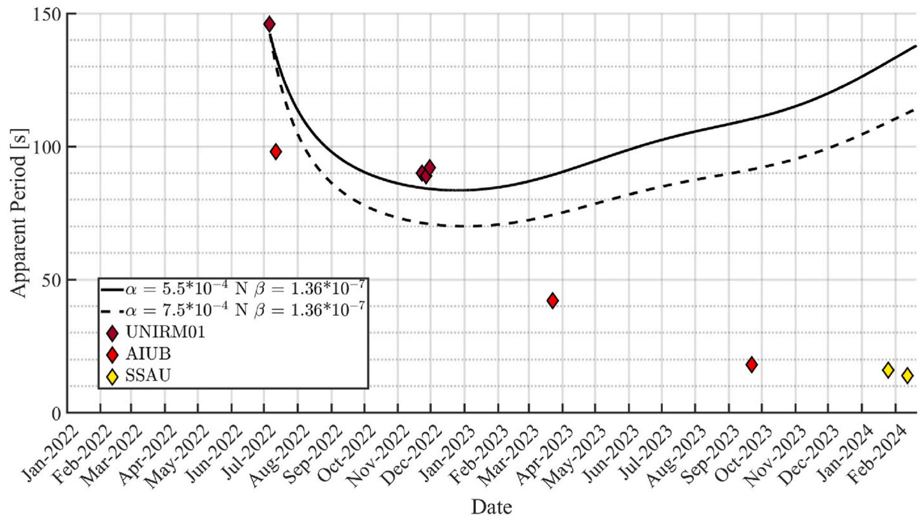

Fig. 7. Apparent period for $\beta = 1.36{\cdot}10^{-7}$.

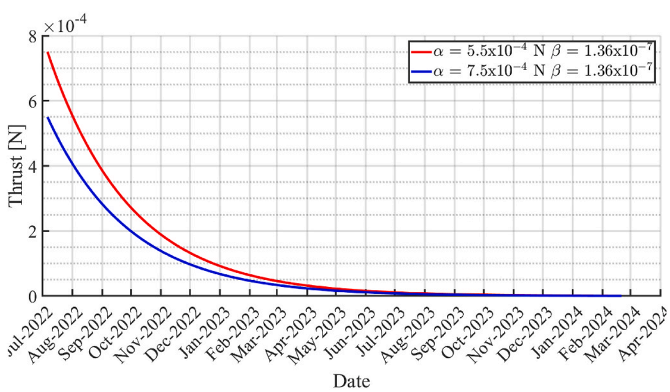

Fig. 8. Thrust profile for $\beta = 1.36{\cdot}10^{-7}$.

UDMH and 1490 L of NTO. At the end of the operation, once all the liquid has been combusted, the system is assumed to be pressurized with nitrogen gas at 3–5 bar. The estimated gas mass is 6–13 kg of nitrogen in the UDMH tank and 7–15 kg in the NTO tank, providing a total impulse of 4000–8500 Ns and 4500–10000 Ns, respectively. Assuming a failure in the second tank and a gas temperature of 250K, with a simulation timespan of 51.4 million seconds, the average thrust is calculated to be 1-2$\cdot10^{-4}$ *N*.

The order of magnitude of the thrust intensity used in the simulation is coherent with these calculations. However, the magnitude of the thrust obtained from the simulation is higher than what would be expected from the pressurizing gas of the RD-861 engine. This difference in

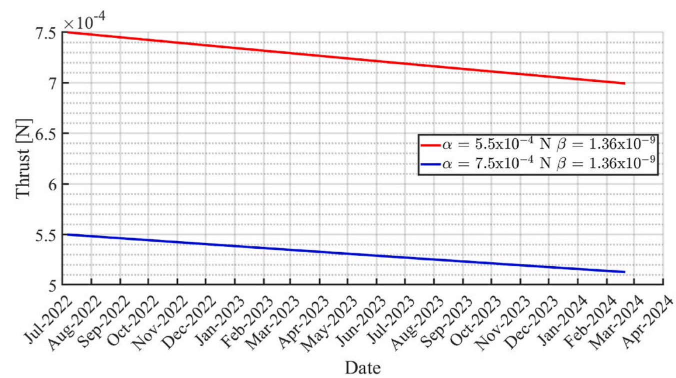

Fig. 10. Thrust profile for $\beta = 1.36{\cdot}10^{-9}$.

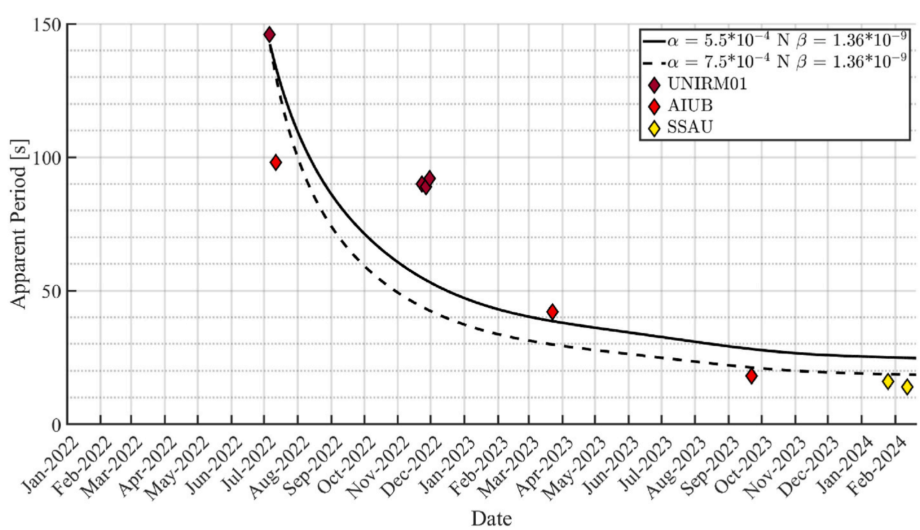

Fig. 9. Apparent period for $\beta = 1.36{\cdot}10^{-9}$.

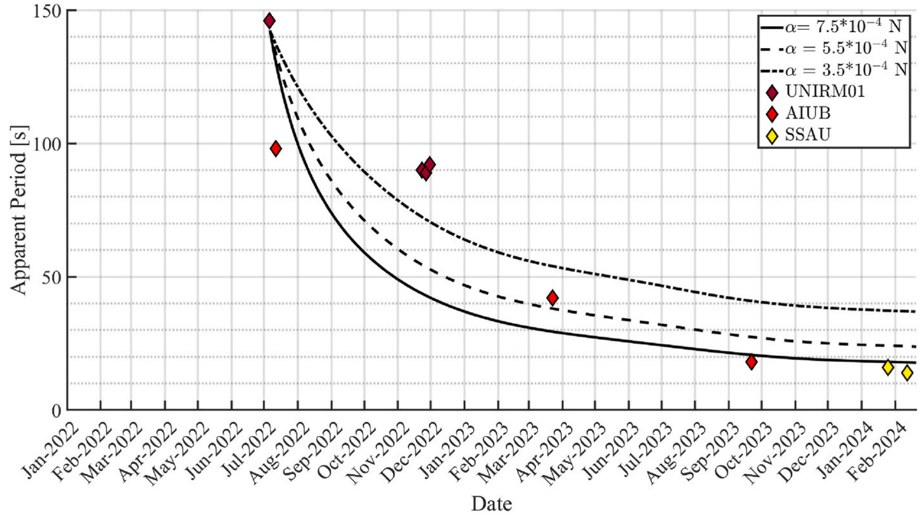

Fig. 11. SL-14 R/B (NORAD ID 18340) attitude simulation from July 2022 to May 2024 with continuous jet leakage perturbation for different thrust values.

the thrust magnitude can be attributed, in a first approximation, to discrepancies between the CAD model used to estimate the system's inertia and the actual physical system. Moreover, as previously noted, the simulated apparent period differs slightly from the observed apparent period.

Additionally, the physical properties of the SL-14 R/B were estimated to best fit the publicly available data; this includes the material properties used in developing the eddy currents perturbation model. The accuracy of the simulated eddy current effects strongly depends on the electrical conductivity, permeability, and structural composition of the SL-14 R/B, all inferred from general assumptions about the materials typically used in similar R/Bs. Any differences between these parameters and the SL-14 R/B's actual properties could lead to variations in the estimated damping effects, potentially affecting the overall attitude evolution.

Despite these differences, the developed perturbation model remains consistent with experimental data, successfully capturing the fundamental dynamics of the event. The estimated disturbance thrust responsible for the observed behaviour has been reliably determined, reinforcing the validity of the applied methodology and contributing to a better understanding of the long-term rotational evolution of non-cooperative space RSOs.

Lastly, it is important to emphasize that, since these assumptions presume the depletion of only one engine tank, it is possible that this event may have occurred in the past or could occur again in the future for the other tank. This further highlights the importance of continuing the follow-up observations of SL-14 R/B.

3.4. Complete simulation

The complete simulation, covering the period from September 2020 to January 2025, is presented in Fig. 12. From September 2020 to July 2022, the attitude of the SL-14 R/B was propagated considering only the perturbation model, which accounts for both gravity gradient and eddy current effects. The thrust model was introduced from July 2022 to February 2024, with the continuous thrust model (7.5 $\times 10^{-4}$ N) proving to be the most representative in reproducing the observational data of SL-14 R/B. This model was applied during the acceleration phase observed in the LC data. Furthermore, following the presumed end of the leakage phase in February 2024, the third phase of the motion evolution,

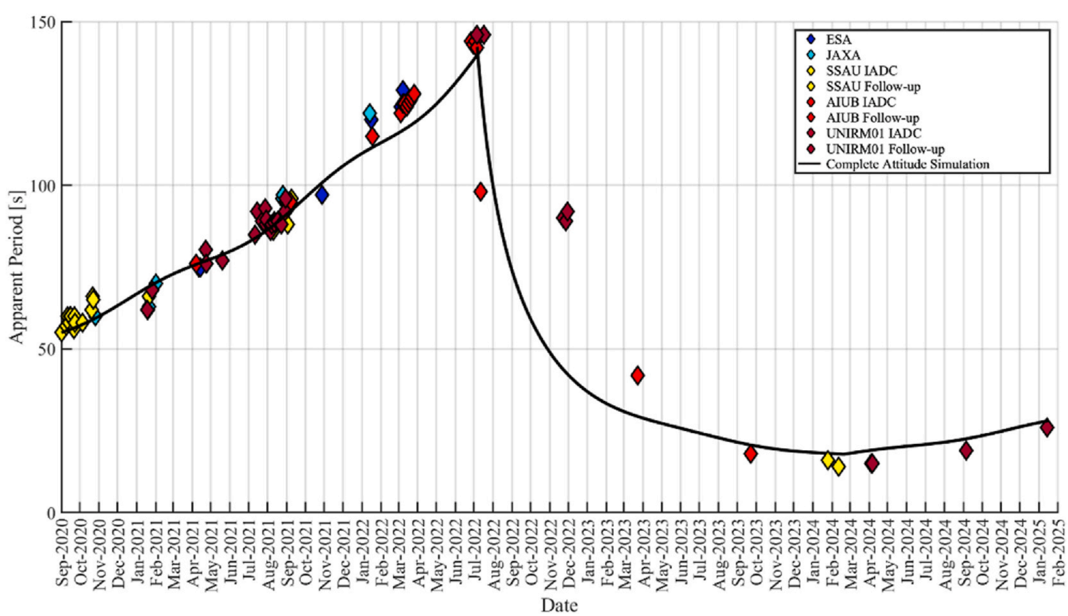

Fig. 12. SL-14 R/B (NORAD ID 18340) complete attitude simulation.

characterized by a deceleration of the SL-14 R/B, was successfully reproduced in the simulation. The agreement between the simulated and observed deceleration reinforces the validity of the perturbation model, confirming its effectiveness in describing the long-term attitude evolution of the SL-14 R/B.

Based on the simulations performed and the available data, the results show that the model successfully reproduces the observed tumbling motion, confirming its effectiveness in describing the evolution of the SL-14 R/B's dynamics. The comparison between the simulated results and the experimental data validates the robustness of the proposed model and its ability to characterize the perturbations affecting the motion of SL-14 R/B.

A more precise characterization of the internal pressure of the tanks, valve mechanics, and gas temperature, which were assumed in this study, would enable a more accurate estimation of the thrust exerted by the RD-861 engine. Additionally, a better understanding of the final state of the propulsion system at the end of its operational life would provide key insights into the exact conditions leading to the presumed leakage. Refining these parameters, together with considering additional perturbation sources would improve the accuracy of the leakage model and its overall impact on the SL-14 R/B's dynamics, thus helping to better define the root causes of the observed anomalous tumbling motion.

4. Digital twins simulations

To study the behaviour of non-cooperative RSOs, computer simulations based on digital twins represent an innovative tool which enables the replication of the observations under different attitude conditions. This approach, combined with real observations and laboratory measurements, is likely to enhance the characterization capabilities regarding these kind of space debris. During this research, a digital twin simulation tool developed at S5Lab, VIrtual Space ObservatioN Engine (VISONE) [20], was used to simulate the pass of the SL-14 R/B above the observatory using different input attitude conditions.

VISONE can be divided in two parts, the first related to orbit and attitude propagation, the second to 3D rendering. Firstly, the RSO and Sun position, as well as the observatory location, are propagated at all instants matching the real observation. The TLE (Two Line Element) [21] provided by NORAD enables the orbit propagation using an SGP4 propagator, while the DE440 ephemeris [22] provided by NASA can be used to retrieve the Sun position. At this stage, also the RSO attitude is propagated, using a free torque model. This approximation is crucial to reduce computation time and does not affect significantly the simulation, since the orbit arc covered is typically very small (duration less than 1 min) and, consequently, not impacted by the external torques.

The second part uses as input the data described above to reconstruct the observation geometry, as well as a CAD model of the investigated RSO. Then, a 3D rendering is performed to produce simulated observation frames. After that, the light reflected by the RSO is computed and at the end of the simulation its synthetic LC can be reconstructed.

By varying the input initial attitude conditions, it is possible to generate several simulated LC, that can be compared to the real observed one. This process, known as LC inversion, leads in principle to the reconstruction of an attitude compatible with the real one. This routine is governed by a Muti-Objective Genetic Algorithm (MOGA) which generates a population of individuals associated to different input attitude conditions to compute synthetic LC and match the real one, obtaining the so-called best individual, i.e. the best compatible attitude. The key advantage of MOGA is the concurrent evaluation of multiple cost functions, which enable considering different properties of the recorded signal. MOGA was implemented using the Python package DEAP [23]. A schematic overview of VISONE is provided in Fig. 13.

The described software has been exploited to attempt the attitude reconstruction of SL-14 R/B. As a notable case, its LC obtained from SCUDO on May 6th, 2024 has been passed as input to VISONE. This case deserved special attention since it is related to the period in which the observed period was at its minimum, indicating a fast-tumbling motion of the SL-14 R/B. Fig. 14 illustrates the result of the LC inversion process, whereas in Table 4 the initial attitude conditions determined by the optimizer can be found. The Root Mean Square (RMS) of the residuals between the actual and simulated data is 2.192 and is considered as the modelling error.

The LC are not perfectly matched. In fact, due to the high complexity

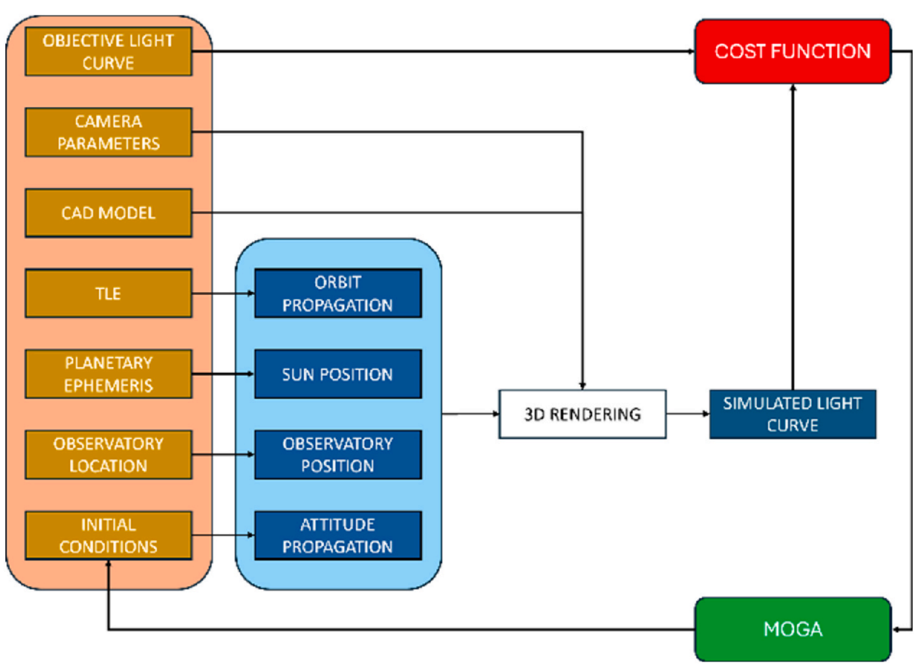

Fig. 13. Overview of VISONE main routines. The left box presents the main inputs. The blue box describes the preliminary processing of the input, necessary to feed the 3D renderer. Finally, the simulated LC and the real one are compared, and a cost function is evaluated by the MOGA, which ultimately adjusts the input attitude conditions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

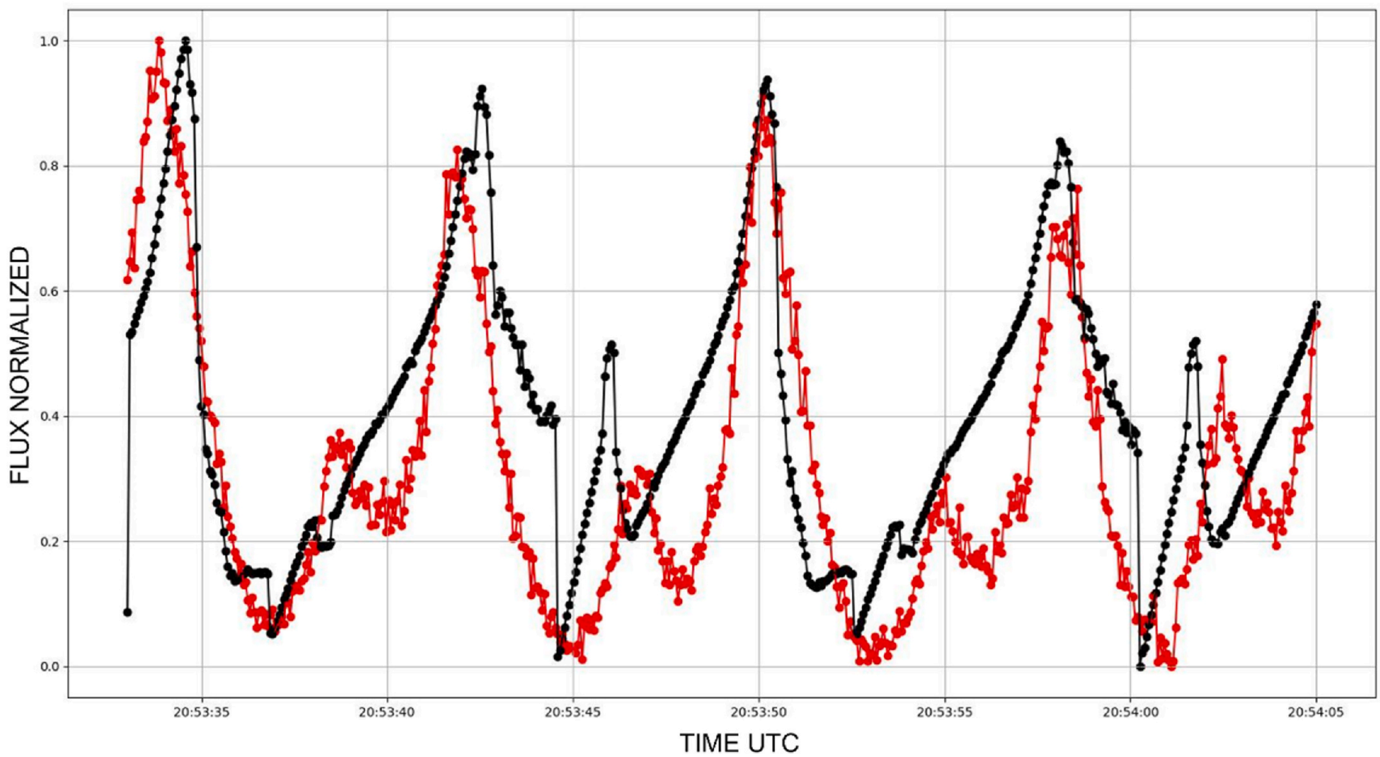

Fig. 14. Comparison between real and simulated LC. The best simulated LC (black) is the curve that better matches the real one (red) and is retrieved starting from a specific initial attitude condition leading to an RMS of 2.192. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 4
Initial quaternions and proper angular velocities of the body which led to the best simulated LC. The first parameter is the scalar part of the quaternion.

q0	q1	q2	q3	p [deg/s]	q [deg/s]	r [deg/s]
0.5498	0.4421	−0.434	0.5063	1.312	0.745	20.1

of the problem many error sources exist, especially when modelling the size, shape and optical properties of the RSO, which directly affect its reflection function. Also, approximations are inevitable when propagating the relative orbit geometry and the RSO attitude. It is also worth noting that this problem is deeply influenced by the tuning of the genetic algorithm parameters. In conclusion, limitations are caused mainly by mathematical approximations, required to speed up the process, and by RSO modelling. However, as illustrated in Fig. 14, obtaining a good estimate of the SL-14 R/B attitude motion is still possible, making the LC inversion and consequently the optical observations a promising tool to sustain research in the field of space surveillance.

Finally, the initial attitude reported in Table 4 was propagated for ten apparent periods to assess the nutation angle of the SL-14 R/B, illustrated in Fig. 15. The nutation angle was considered as the angle between the object angular velocity vector and its angular momentum, both resolved in the object body reference frame, as illustrated in the sketch provided in Fig. 16. Additionally, to understand the SL-14 R/B's orientation with respect to the orbital plane, the angle between the normal to the orbital plane, which is the orbital angular momentum vector $\widehat{h}$, and the minor axis of inertia (the one aligned with the cylindrical shape denoted as vector $\widehat{b}_3$) was evaluated. This is illustrated in Figs. 17 and 18. In the latter figure, a picture of the SL-14 R/B's orientation relative to its orbit is shown.

Additionally, to understand the orientation of the SL-14 R/B with respect to the orbital plane, the angle between the normal to the orbital plane, i.e. the orbital angular momentum vector $\widehat{h}$, and the minor axis of inertia $\widehat{b}_3$ was evaluated. The $\widehat{b}_3$ axis is rigidly attached to the body of the SL-14 R/B and aligned with its cylindrical geometry. This analysis is illustrated in Figs. 17 and 18, the latter showing a visual representation of the SL-14 R/B's orientation relative to its orbit.

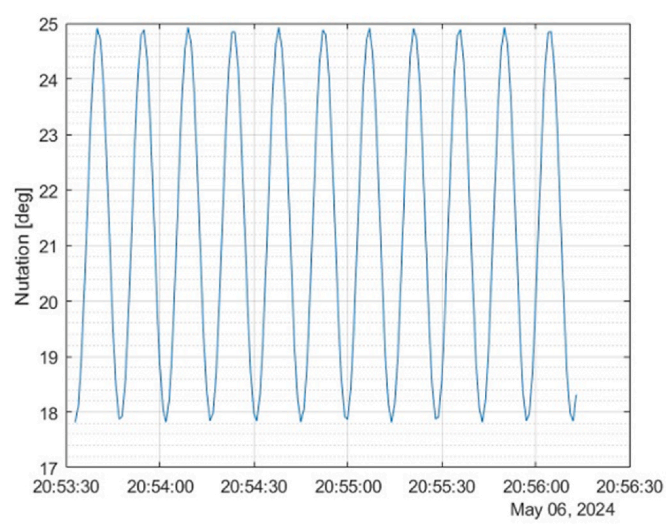

Fig. 15. Nutation angle during ten apparent periods (approximately 160 s).

5. Conclusions and future work

This study demonstrates the effectiveness of Light Curves (LC) analysis as a tool for detecting and analysing the tumbling motion of Resident Space Objects (RSO), contributing to a better understanding of

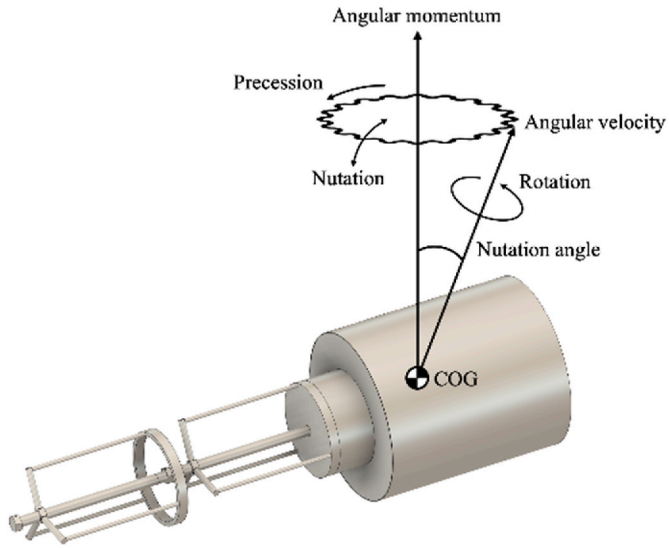

Fig. 16. Nutation angle of SL-14 R/B (NORAD ID 18340).

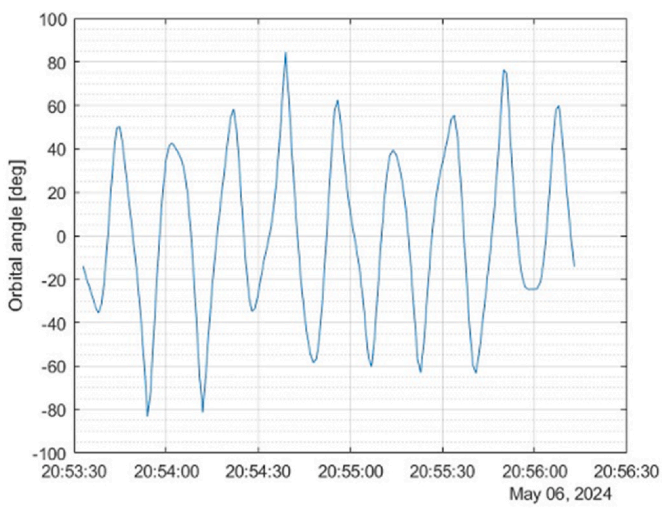

Fig. 17. Angle between the orbital plane and the minor axis of inertia.

uncontrolled orbiting RSOs and supporting Space Surveillance and Tracking (SST) activities. The Light Curve Sequential Comparison Strategy, introduced in this paper, has proven essential in monitoring the long-term evolution of tumbling RSOs, providing a valuable dataset for tracking attitude variations over extended periods.

Through international collaboration within the Inter-Agency Space Debris Coordination Committee (IADC), S5Lab has compiled a historical record of LC data, enabling the identification of an anomalous rotational behaviour in SL-14 R/B (NORAD ID 18340). The sequential LC analysis revealed a sudden acceleration phase, prompting further investigations into the underlying physical mechanisms responsible for this unexpected change in rotational state.

To go beyond simple tumbling detection, attitude simulations were conducted, integrating both environmental perturbation models (gravity gradient and eddy currents) and possible internal disturbances. The results validated the perturbation model and confirmed that the observed acceleration phase could be attributed to a mechanical failure in the propulsion system, leading to a pressurized gas leakage from one of the attitude control nozzles. Two different thrust models (exponential decay and continuous leakage) were tested, with the latter providing a closer match to the observational data, reinforcing the plausibility of the failure hypothesis. While the order of magnitude of the simulated thrust

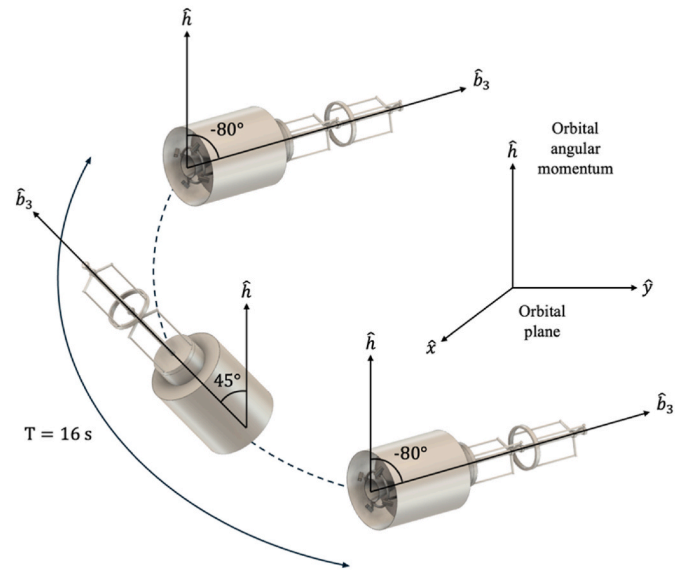

Fig. 18. Orientation of SL-14 R/B (NORAD ID 18340) during a single apparent rotation period as computed by propagating the initial attitude.

is consistent with what can be expected from a probable pressurizing gas leakage, discrepancies in the thrust magnitude could arise due to uncertainties in the CAD-based inertia estimation and the approximations in the leakage model. These differences suggest that further refinements, such as improving the modelling of the internal pressure distribution, valve dynamics, and gas temperature, are necessary to achieve a more precise determination of the actual thrust profile. Moreover, obtaining more detailed information on the RD-861 engine, particularly regarding its final state at the end of operations, would be fundamental in accurately assessing the real gas leakage thrust value. This would allow for a more rigorous validation of the leakage hypothesis and an improved characterization of the SL-14 R/B's attitude evolution.

Beyond attitude simulations, this work also highlights the potential of digital twin simulations in RSO characterization. Using VIrtual Space ObservatioN Engine (VISONE), a digital twin approach was applied to reconstruct the observed light curve and estimate SL-14 R/B's attitude motion. Despite some mismatches due to inevitable modelling uncertainties, the results demonstrate that LC inversion techniques provide a promising tool for space surveillance applications, offering new avenues for the study of non-cooperative RSOs.

In conclusion this study highlights the importance of international cooperation and data sharing in advancing SST. Strengthening collaboration among space agencies and international committees, such as the IADC and the United Nations Committee on the Peaceful Uses of Outer Space (UN-COPUOS), will enhance predictive models and RSO characterization, leading to more accurate assessments of space debris dynamics. Moreover, the findings presented provide new insights into the behaviour of non-cooperative RSOs, emphasizing the need for continuous observation and advanced modelling techniques to ensure the safe and sustainable use of Earth's orbital environment. These advancements will support space situational awareness, space traffic management, and debris mitigation strategies, contributing to the protection of operational satellites and the long-term sustainability of space activities.

CRediT authorship contribution statement

S. Kumar: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Data curation. **L. Chiavari:** Writing – review & editing, Writing – original draft, Software, Project administration, Data curation. **L. Cimino:** Writing – review & editing, Validation, Software. **S. Varanese:** Writing – review & editing, Supervision, Software, Methodology, Data curation. **L. Mariani:** Supervision. **M. Rossetti:** Supervision, Data curation. **G. Zarcone:** Supervision. **T. Schildknecht:** Writing – review & editing, Supervision, Conceptualization. **F. Nasuti:** Writing – review & editing, Supervision, Methodology, Data curation. **F. Piergentili:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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